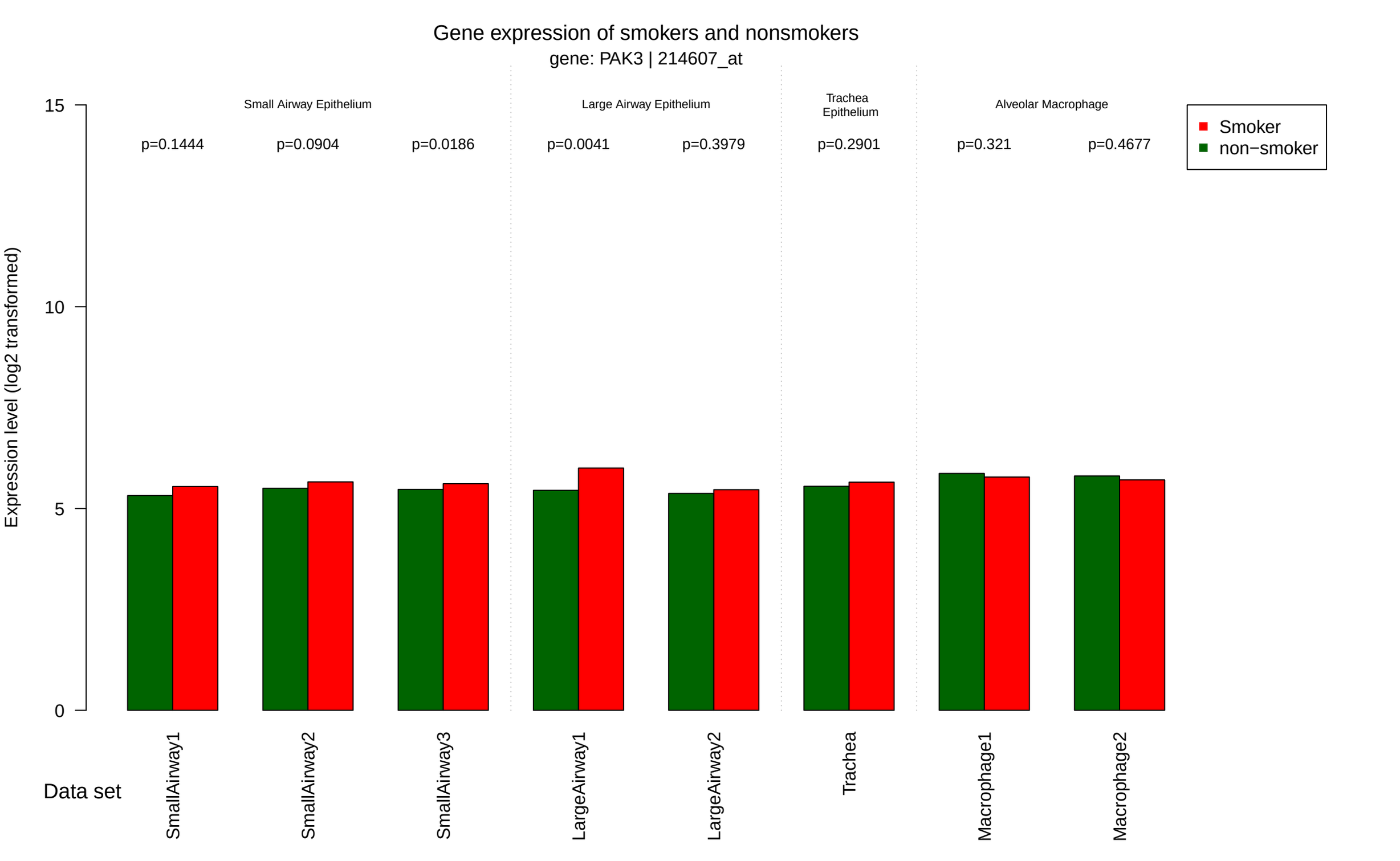
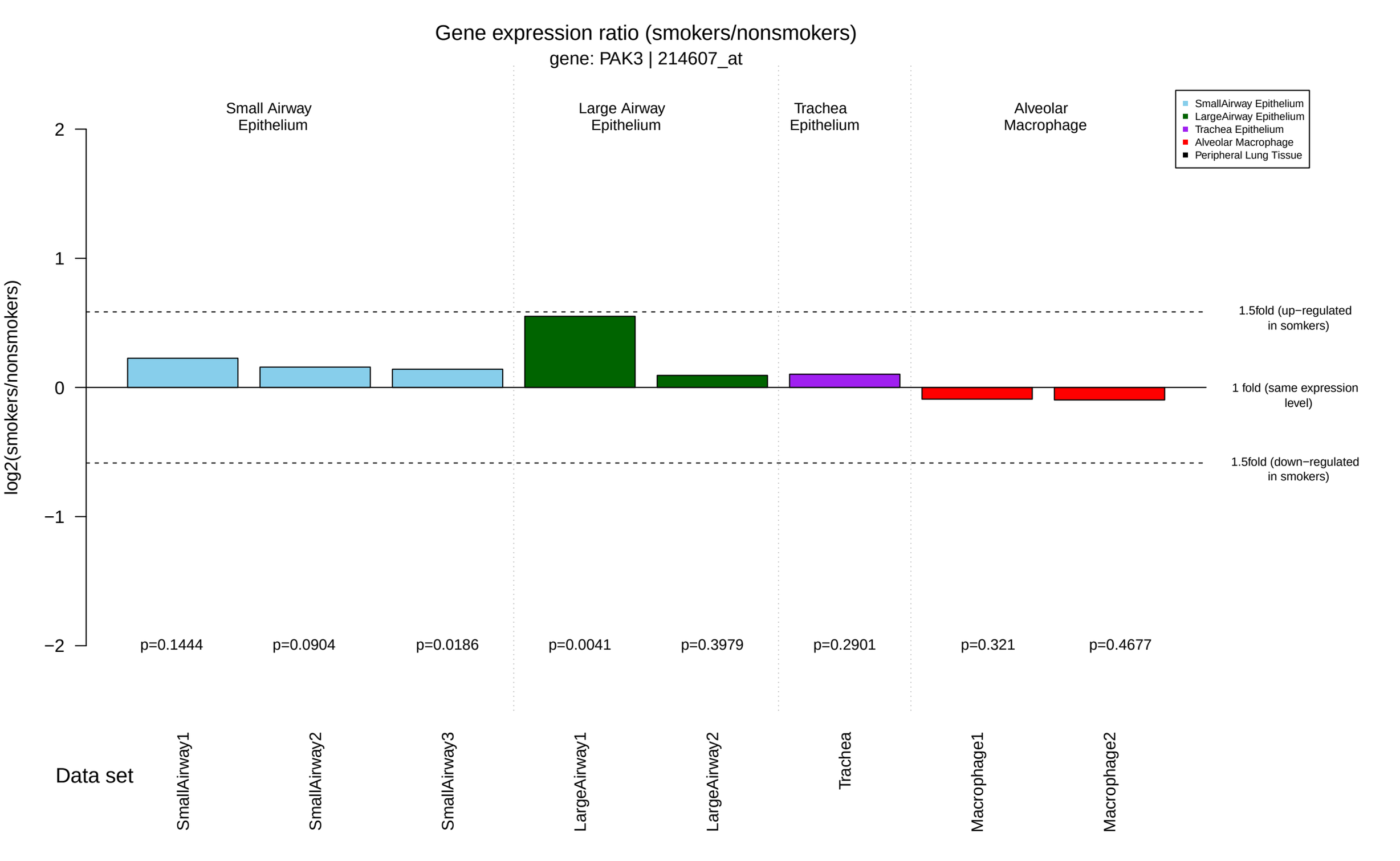
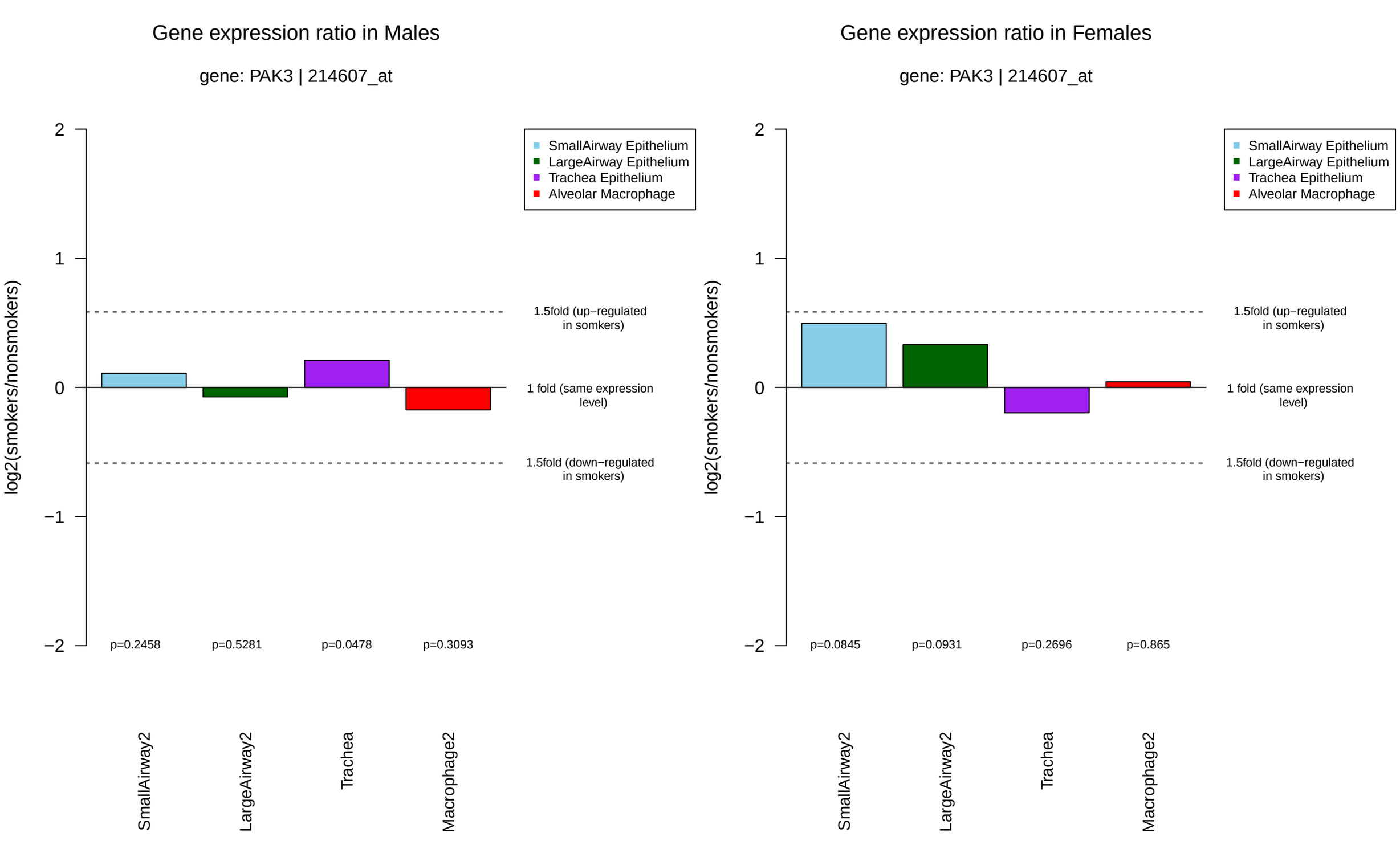
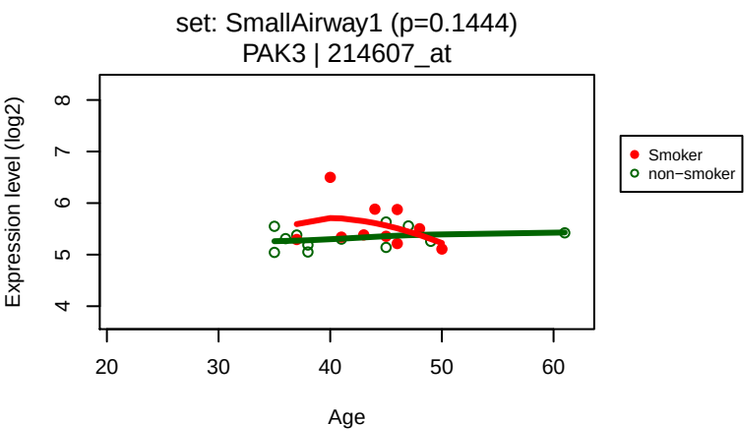
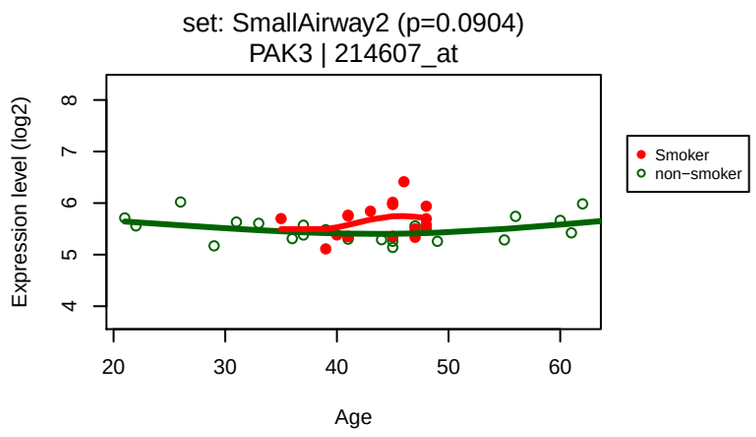
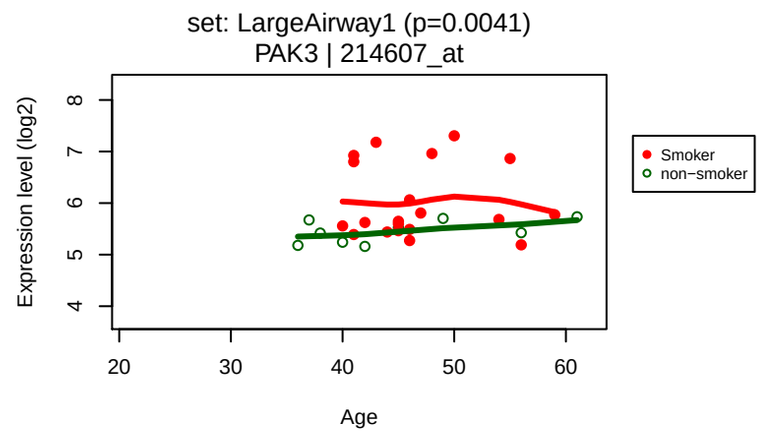

Gene expression of smokers and nonsmokers
gene: PAK3 | 214607_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1444
p=0.0904
p=0.0186
p=0.0041
p=0.3979
p=0.2901
p=0.321
p=0.4677
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

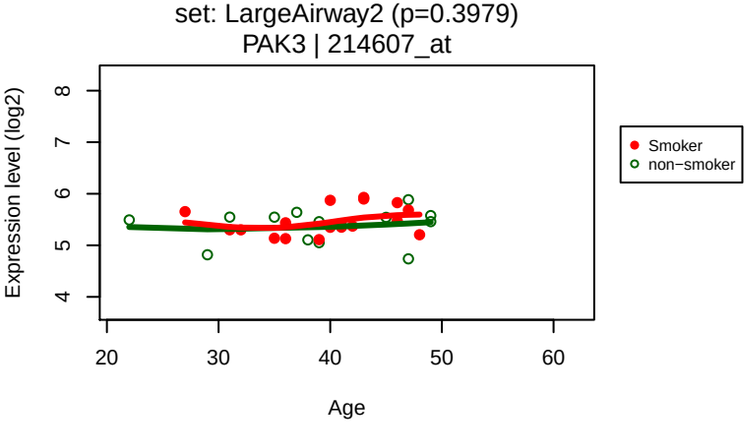

Gene expression ratio (smokers/nonsmokers)
gene: PAK3 | 214607_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1444
p=0.0904
p=0.0186
p=0.0041
p=0.3979
p=0.2901
p=0.321
p=0.4677
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

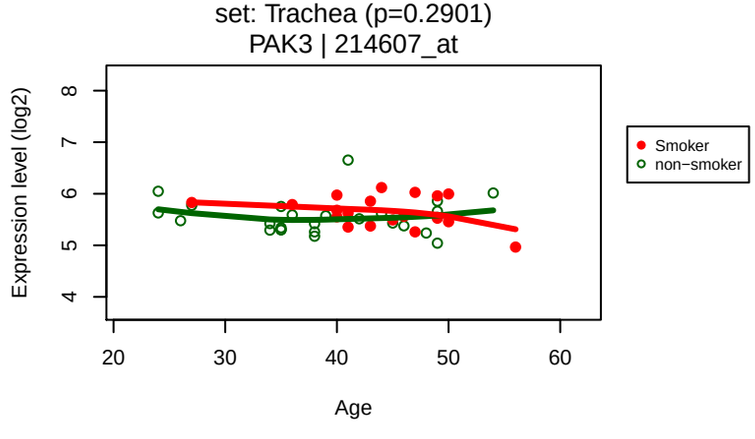

Gene expression ratio in Males
gene: PAK3 | 214607_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2458
p=0.5281
p=0.0478
p=0.3093
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PAK3 | 214607_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0845
p=0.0931
p=0.2696
p=0.865
SmallAirway2
LargeAirway2
Trachea
Macrophage2

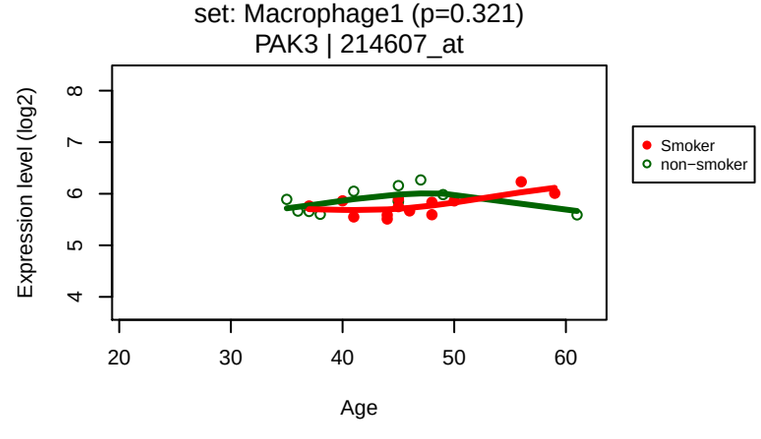

set: SmallAirway1 (p=0.1444)
PAK3 | 214607_at
Expression level (log2)
Age
Smoker
non−smoker

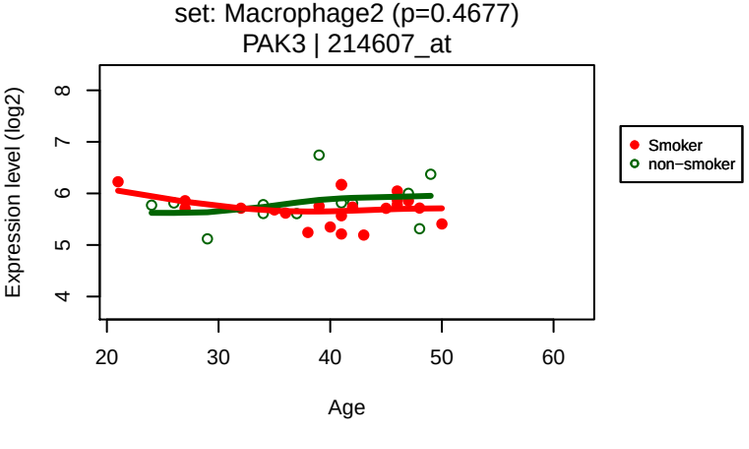

set: SmallAirway2 (p=0.0904)
PAK3 | 214607_at
Expression level (log2)
Age
Smoker
non−smoker

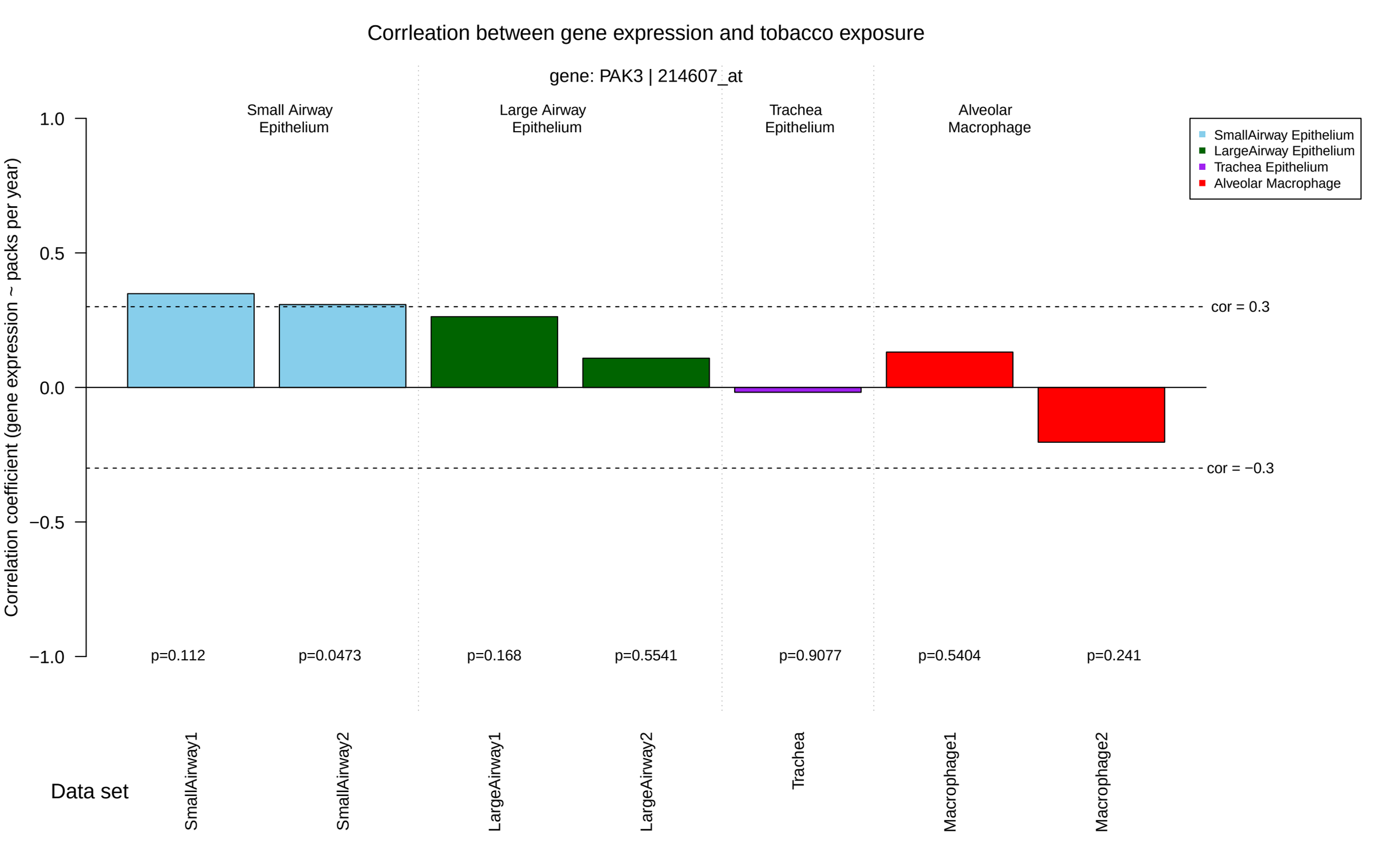

set: LargeAirway1 (p=0.0041)
PAK3 | 214607_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3979)
PAK3 | 214607_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2901)
PAK3 | 214607_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.321)
PAK3 | 214607_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4677)
PAK3 | 214607_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PAK3 | 214607_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.112
p=0.0473
p=0.168
p=0.5541
p=0.9077
p=0.5404
p=0.241
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2